

Ali Raza

Full-Stack AI Engineer

Multan, Pakistan | +92 324 5892281 | aliraza.ai.dev3@gmail.com
linkedin.com/in/ali-raza-597b01360 | github.com/AliRaza-Dev678

Professional Summary

Full-Stack AI Engineer and Software Engineering undergraduate building intelligent products across React/TypeScript frontends, FastAPI services, PostgreSQL/Redis data layers, LLM workflows, and n8n automation pipelines. Designed a containerized AI-powered workspace platform with secure authentication, role-aware access, real-time WebSocket communication, and Groq-powered insights. Additional experience includes document and video RAG, LangGraph agents, NLP, and computer vision, with a focus on turning AI prototypes into complete, usable software systems.

Technical Skills

- Full-Stack & Platform Engineering:** Python, FastAPI, React, TypeScript, JavaScript, Vite, REST APIs, WebSockets, JWT/OAuth2, role-aware authorization, Pydantic
- Generative AI & LLMs:** RAG pipelines, agentic systems, prompt engineering, embeddings, semantic search, MMR retrieval, LangChain, LangGraph, ReAct, Groq API, Hugging Face, Llama models
- Data, Persistence & Background Processing:** PostgreSQL, PGVector, Redis, Celery, Tortoise ORM, Aerich, SQLAlchemy, SQLite
- Machine Learning, NLP & Vision:** Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, TF-IDF, sentiment analysis, classification, clustering, CNNs, LSTMs, transfer learning, ResNet50
- Workflow Automation & Deployment:** n8n, Docker Compose, Streamlit, Git, GitHub, Jupyter Notebook, Google Colab, OpenAPI/Swagger

Selected Projects

Intelligent AI Assistant & Content Automation Pipelines *n8n, Groq, LangChain, Google Workspace APIs, Meta API*

- Engineered a Personal Assistant Agent within n8n, utilizing Groq and LangChain to automate Google Calendar scheduling, Gmail management, Google Docs generation, and Google Sheets expense tracking.
- Developed a modular content distribution engine with n8n, polling Google Sheets for state management and automatically publishing media to YouTube and Instagram Reels via the Meta Graph API.
- Implemented asynchronous branching, parallel execution tracks, and web search integration for reliable multi-step agentic workflows.

AI-Powered Workspace Management Platform *Python, FastAPI, React, PostgreSQL, Redis, Celery, Docker*

- Designed a full-stack collaboration platform for workspaces, teams, projects, tasks, files, and analytics.
- Developed secure JWT authentication, role-aware authorization, real-time WebSockets, and Groq-powered insights.
- Added Redis caching, Celery background reports, CSV exports; containerized services with Docker Compose.

AI Chat Assistant (ChatGPT Clone) *React, TypeScript, FastAPI, LangGraph, PostgreSQL, Docker*

- Built a production-style assistant with React/TypeScript, FastAPI, and LangGraph serving Groq-hosted gpt-oss-120b.
- Implemented token-by-token streaming, persistent conversations, stop/regenerate controls, and Markdown rendering.

IUB University & YouTube RAG Chatbots *LangChain, PGVector, PostgreSQL, Hugging Face, Groq, Streamlit*

- Built RAG assistants to ingest IUB PDFs and YouTube transcripts (via yt-dlp), storing vectors in PGVector.
- Orchestrated grounded answer generation with Llama 3.1 via Groq and MMR retrieval using Streamlit interfaces.

Professional Experience

Machine Learning Engineer Intern | *Developers Hub Corporation (Remote)* **2025**

Built an end-to-end IMDB sentiment system using TF-IDF features, linear classifiers, and a Streamlit interface. Developed ML workflows for NLP, classification, and clustering tasks with reproducible prediction pipelines.

Machine Learning Engineer Intern | *Code Alpha (Remote)* **2025**

Completed projects in heart-disease prediction, agricultural image recognition, and speech-emotion recognition. Trained logistic-regression, CNN, and LSTM models with structured preprocessing and performance analysis.

Education

Bachelor of Science in Software Engineering | *The Islamia University of Bahawalpur* **2023–2027**

FSc Pre-Engineering | *KIPS College* **2021–2023**

Professional Development & Continuous Learning

- Currently Learning:** GHL (GoHighLevel) for marketing automation and CRM integration.
- Generative AI & LLM Engineering:** CampusX: LangChain, RAG pipelines, ReAct agent development.
- Machine Learning & Deep Learning:** Coursework through Hope to Skills and Sidhardan.
- Backend Engineering:** FastAPI, Tortoise ORM, JWT, PostgreSQL, Redis, Celery, Docker.